

INITIATOR A

RESPONDER B

Phase 1: Handshake Initiation & Preparation

1. $(pk_{eph}^A, sk_{eph}^A) \xleftarrow{\$} \text{K.KeyGen}()$
2. $fp_A \leftarrow \text{SHA3-256}(pk_{eph}^A)$
3. $N_A, token_{raw} \xleftarrow{\$} \{0, 1\}^{256}$
4. $(K_{static}, C_{static}) \leftarrow \text{K.Encaps}(pk_{Kyber}^B)$

Symmetric Binding & RSA KEM Layer

5. $k_{verify} \leftarrow \text{HKDF}(\text{"DOS.SHIELD"} \parallel K_{static} \parallel N_A \parallel ID_A)$
6. $MAC_1 \leftarrow \text{HMAC}(k_{verify}, ID_A \parallel pk_{eph}^A \parallel N_A \parallel C_{static})$
7. $AD_1 \leftarrow ID_A \parallel N_A \parallel MAC_1$
8. $(k_{RSA}^A, C_{RSA}^A) \leftarrow \text{R.Encaps}(pk_{RSA}^B)$
9. $token_A \leftarrow ID_A \parallel fp_A \parallel N_A \parallel token_{raw}$
10. $C_{token} \leftarrow \text{AEAD.Seal}(k_{RSA}^A, token_A, AD_1)$

$M1 = \langle ID_A, pk_{eph}^A, C_{static}, C_{RSA}^A, C_{token}, N_A, MAC_1 \rangle$

[LAYER 1] Early Anti-DoS Shield (Fast)

11. $K_{static} \leftarrow \text{K.Decaps}(C_{static}, sk_{Kyber}^B)$
12. $k_{verify} \leftarrow \text{HKDF}(\text{"DOS.SHIELD"} \parallel K_{static} \parallel N_A \parallel ID_A)$
13. $expected_MAC_1 \leftarrow \text{HMAC}(k_{verify}, ID_A \parallel pk_{eph}^A \parallel N_A \parallel C_{static})$
14. **Verify** $MAC_1 = expected_MAC_1$
If invalid $\rightarrow$ Abort immediately (Drop early)

[LAYER 2] Core Cryptographic Engine (Heavy)

15. $k_{RSA}^A \leftarrow \text{R.Decaps}(C_{RSA}^A, sk_{RSA}^B)$
16. $token_A \leftarrow \text{AEAD.Open}(k_{RSA}^A, C_{token}, AD_1)$
17. **Parse** $token_A \rightarrow (ID'_A, fp'_A, N'_A, token_{raw})$
18. **Verify** $ID'_A = ID_A, N'_A = N_A, fp'_A = \text{SHA3-256}(pk_{eph}^A)$
19. $H_1 \leftarrow \text{SHA3-256}(M1); N_B \xleftarrow{\$} \{0, 1\}^{256}$
20. $(K_{eph}, C_{eph}) \leftarrow \text{K.Encaps}(pk_{eph}^A)$
21. $(k_{RSA}^B, C_{RSA}^B) \leftarrow \text{R.Encaps}(pk_{RSA}^A)$
22. $H_2 \leftarrow \text{SHA3-256}(H_1 \parallel C_{eph} \parallel C_{RSA}^B \parallel N_B)$

$M2 = \langle C_{eph}, C_{RSA}^B, C_{challenge}, N_B \rangle$

Phase 3: Client Verification

26. $H_2 \leftarrow \text{SHA3-256}(H_1 \parallel C_{eph} \parallel C_{RSA}^B \parallel N_B)$
27. $K_{eph} \leftarrow \text{K.Decaps}(C_{eph}, sk_{eph}^A)$
28. $K_{master} \leftarrow \text{HKDF}(K_{static} \parallel K_{eph} \parallel H_2)$
29. $k_{RSA}^B \leftarrow \text{R.Decaps}(C_{RSA}^B, sk_{RSA}^A)$
30. $challenge \leftarrow \text{AEAD.Open}(k_{RSA}^B, C_{challenge}, AD = H_2)$
31. **Verify** $challenge = \text{HMAC}(K_{master}, \text{"challenge"} \parallel H_2 \parallel token_{raw})$
32. $response \leftarrow \text{HMAC}(K_{master}, \text{"response"} \parallel H_2 \parallel challenge)$

$M3 = \langle response \rangle$

Transcript Binding & Challenge Construction

23. $K_{master} \leftarrow \text{HKDF}(K_{static} \parallel K_{eph} \parallel H_2)$
24. $challenge \leftarrow \text{HMAC}(K_{master}, \text{"challenge"} \parallel H_2 \parallel token_{raw})$
25. $C_{challenge} \leftarrow \text{AEAD.Seal}(k_{RSA}^B, challenge, AD = H_2)$

$K_{session} \leftarrow \text{HKDF}(\text{"SESSION"} \parallel K_{master} \parallel H_2)$

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